

Due 9/21/2026 by 9:05 am

**1. Sector Selection & Data Acquisition**

- Select **two** S&P 500 sectors from the provided list that have at least 20 companies in the Kept column.
- Load the corresponding CSV file(s) containing company ticker-CIK pairs.
- Extract primary financial concepts via SEC EDGAR company facts API across the 2020–2025 fiscal years:
  - Net Income (NetIncomeLoss)
  - Revenue (RevenueFromContractWithCustomerExcludingAssessedTax or SalesRevenueNet)
  - Total Assets (Assets)
  - Stockholders' Equity (StockholdersEquity)
- Compute the 3-Step DuPont Components:

$$ROE = \left( \frac{\text{Net Income}}{\text{Revenue}} \right) \times \left( \frac{\text{Revenue}}{\text{Total Assets}} \right) \times \left( \frac{\text{Total Assets}}{\text{Stockholders' Equity}} \right)$$

- **Data Hygiene Protocol:** You may drop companies that yield unresolvable NaN values or missing filings. Your final usable company count per sector should align with the provided Kept column counts.

| Sector                 | Count | Kept | Dropped | Kept% |
|------------------------|-------|------|---------|-------|
| Industrials ✖          | 83    | 58   | 25      | 69    |
| Information Technology | 73    | 57   | 16      | 78    |
| Health Care ✖          | 59    | 45   | 14      | 76    |
| Consumer Discretionary | 47    | 38   | 9       | 81    |
| Consumer Staples       | 34    | 24   | 10      | 71    |
| Utilities              | 31    | 12   | 19      | 32    |
| Real Estate            | 30    | 21   | 9       | 70    |
| Materials              | 25    | 14   | 11      | 56    |
| Communication Services | 24    | 15   | 9       | 62    |
| Energy                 | 21    | 16   | 5       | 72    |

**3. Cross-Sector Analysis**

- Address the following analytical questions in your report:
  - How do the sectors that you selected differ in their primary engine of return (e.g., High Profit Margins vs. High Asset Turnover vs. High Financial Leverage)?
  - What macro trends over 2020–2025 (e.g., supply chain shifts, interest rate changes, tech demand) are reflected in the sub-component trajectories?
- Support your findings with appropriate metrics and visualizations.

**4. Company-Level Outlier Analysis**

- Apply a statistical technique of your choice to identify company outliers within each sector. (Note that we are looking for companies that are outliers, not specific company and year combinations.)
- Explain your chosen methodology and document which specific metrics (ROE, Profit Margin, Asset Turnover, or Financial Leverage) are likely to have triggered the outlier designation.

## DA 401 Homework #4

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### 5. Single-Year Event Anomaly Detection

- Identify specific **Company-Year** pair anomalies (e.g., a single-year spike in ROE or a sudden collapse in Asset Turnover).
- Investigate the probable underlying cause using 10-K disclosures, news releases, or corporate actions (e.g., major restructuring charges, asset write-downs, windfalls, tax adjustments, or share buybacks).

### 6. Binary Industry Clustering ( $K = 2$ )

- Scale/standardize your average DuPont components and fit a clustering model (using the clustering technique of your choice) specifying  $K = 2$ .
- Evaluate:
  - Does the unsupervised model naturally segregate companies back into their official broad sector categories?
  - If not, what shared financial traits cause cross-industry clusters to form instead?

### 7. Optimal Clustering ( $K^*$ ) & Profiling

- Determine the optimal cluster count ( $K^*$ ) using standard metrics (e.g., Silhouette Score or Elbow Method/Inertia).
- Profile each resulting cluster by describing its defining financial archetype (e.g., "High-Margin Low-Turnover Tech Giants" vs. "Asset-Heavy Low-Margin Industrials") using central tendencies of each feature.